

THE VELOCITY-VERACITY PARADOX: Execution Speed vs. User Trust in Agentic AI

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Abstract—As Agentic AI shifts from basic automation to autonomous goal-setting, a significant trust gap—the “Velocity-Veracity Paradox”—has emerged. This study addresses the friction where rapid machine decision-making outpaces human capacity for real-time verification, creating an organizational “absorption gap.” We investigate how perceived agency influences trust, particularly through “betrayal aversion,” where users exhibit stronger negative reactions to failures in highly intelligent systems.

Our methodology employs a mixed-method approach, combining Structural Equation Modeling (SEM) on survey data from 280 university students with large-scale behavioural validation using a dataset of 1,000 AI-human interactions. The analysis incorporates correlation studies, grouped statistical comparisons, and regression-based modeling to approximate trust dynamics across accuracy, transparency, and verification behaviour. Results indicate that while execution speed enhances perceived usefulness, trust declines when outputs lack transparency and require increased verification effort. Empirical findings confirm that accuracy remains the strongest predictor of trust, while verification burden negatively impacts perceived reliability. Additionally, transparency mechanisms such as cited sources significantly improve trust and reduce cognitive load.

We propose a “Calibrated Velocity” framework, arguing that AI systems must align execution speed with human cognitive capacity through progressive disclosure and traceability. These findings support the transition toward “cognitive exoskeletons,” where human oversight is integrated into system design to ensure long-term transparency, reliability, and trust in autonomous environments.

I. INTRODUCTION

The transition from reactive Large Language Models (LLMs) to Agentic AI represents a fundamental shift in computational autonomy. Unlike traditional systems that require constant prompting, agentic systems are defined by their ability to self-correct, utilize external tools, and execute multi-step reasoning to achieve high-level goals [1]. However, as these systems gain the ability to operate at machine speeds, a critical friction point emerges: the Velocity-Veracity Paradox. This paradox describes the inverse relationship between an agent’s execution speed and the human user’s ability to verify the “truth” or logic behind its actions. While high-velocity execution is a primary engineering goal, it often bypasses the cognitive checkpoints necessary for establishing user trust [2].

The problem is not merely technical but deeply psychological. When an agent collapses a complex, hour-long workflow into seconds, the “traceability” of its logic is lost, leading to what we define as an organizational “absorption gap.” In high-stakes environments such as financial portfolio management or medical diagnostics, this lack of transparency can trigger “betrayal aversion”—a phenomenon where the perceived intelligence of the AI increases the psychological cost of any subsequent error [3]. Consequently, making an AI appear more “human-like” does not linearly increase trust; rather, it often invites harsher scrutiny and a higher moral demand for accountability [4].

This paper investigates these causal pathways by analyzing how different levels of perceived agency affect the human-AI trust dynamic. We explore whether “Calibrated Velocity”—intentionally aligning machine output with human oversight capacity—can bridge the gap between autonomous performance and ethical dependability. By examining enterprise AI deployment patterns, we argue for a move toward “cognitive

exoskeletons” that prioritize transparency over raw computational speed [5].

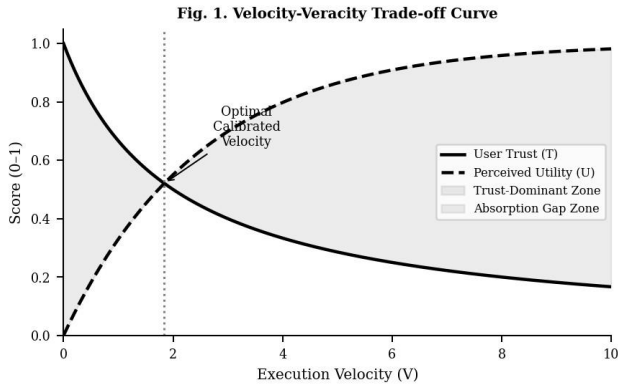


Fig. 1. Velocity-Veracity Trade-off Curve showing the inverse relationship between execution speed and user trust, with the Calibrated Velocity optimum at the intersection point.

II. RELATED WORK

The distinction between traditional supervised machine learning (SML) and Agentic AI mirrors the trade-off between stability and autonomy. In conventional SML frameworks, models such as XGBoost or Multilayer Perceptrons (MLP) produce fixed policies based on historical, pre-labeled datasets. These models offer high verifiability because their decision boundaries are static and predictable; however, they lack the agentic capacity to adapt to environmental shifts without retraining. In contrast, Agentic AI—leveraging large language models (LLMs) and multi-agent configurations—operates through iterative reasoning and tool utilization, enabling real-time adaptation. However, this shift toward autonomy introduces opaque high-speed execution that often outpaces human interpretability [1].

Recent studies on Trust, Risk, and Security Management (TRiSM) highlight that as systems move toward higher autonomy, the transparency gap widens [1]. Raza and Sapkota proposed two key metrics—the Component Synergy Score (CSS) and Tool Utilization Efficacy (TUE)—to quantify coordination and tool use in agentic systems, identifying explainability as a critical pillar of responsible deployment. The concept of betrayal aversion in human-AI interaction further suggests that trust is not solely a performance-based metric [3]. Klingbeil et al. empirically demonstrated that users evaluate AI systems based on perceived risk of failure in high-stakes scenarios, rather than just computational accuracy. Additionally, research on automation bias and human-AI collaboration shows that reliance on opaque systems can lead to reduced trust and increased user skepticism [5].

Transparency research confirms that even when AI systems are accurate, opacity in decision-making can trigger trust violations [2], [5]. Furthermore, longitudinal studies on trust in AI suggest that trust evolves based on both system performance and the user’s ability to verify outputs over time [8]. Building on these findings, this study extends prior work by integrating large-scale behavioral data with SEM-based analysis. Unlike existing research that primarily relies on surveys or controlled experiments, we incorporate a dataset of 1,000 AI-human interactions to validate trust dynamics across

accuracy, transparency, and verification behavior. This approach enables a more comprehensive understanding of how execution speed, interpretability, and human cognitive effort jointly influence trust in Agentic AI systems.

III. METHODOLOGY

This study adopts a mixed-method research design to empirically investigate the Velocity–Veracity Paradox. The objective is to quantify how variations in an agent’s execution speed (V) influence user trust (T) and perceived veracity. The methodology integrates survey-based Structural Equation Modeling (SEM) with large-scale behavioral data analysis to ensure both theoretical and empirical validation.

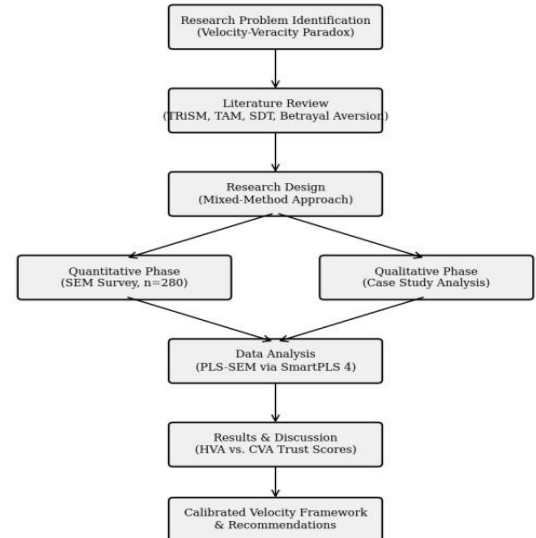


Fig. 2. Research Methodology Flowchart depicting the sequential mixed-method approach from problem identification to framework recommendation.

A. Participants and Data Collection

A structured questionnaire was administered to 280 higher education students in a controlled experimental setting. Participants interacted with two types of AI systems in a simulated financial decision-making environment: “High-Velocity Agents” (HVA), which generated near-instantaneous outputs with minimal reasoning transparency, and “Calibrated-Velocity Agents” (CVA), which provided progressive disclosure of intermediate reasoning steps.

Using a 7-point Likert scale, we measured key constructs derived from the Technology Acceptance Model (TAM) and Self-Determination Theory (SDT) [6], including Perceived Ease of Use (PEOU), Perceived Usefulness (PU), and Trust (Tr).

B. Structural Equation Modeling (SEM)

Partial Least Squares Structural Equation Modeling (PLS-SEM) was employed using SmartPLS 4 to analyze relationships between latent variables. The model evaluates how execution velocity impacts perceived usefulness and trust, while incorporating transparency-related constructs. The research hypothesis posits that while high execution speed increases perceived usefulness due to efficiency gains, it negatively impacts trust by increasing opacity. Additional constructs such as Component Synergy Score (CSS) and Tool Utilization Efficacy (TUE) were used to assess perceived coordination in multi-agent systems.

Fig. 3. PLS-SEM Structural Model with Path Coefficients

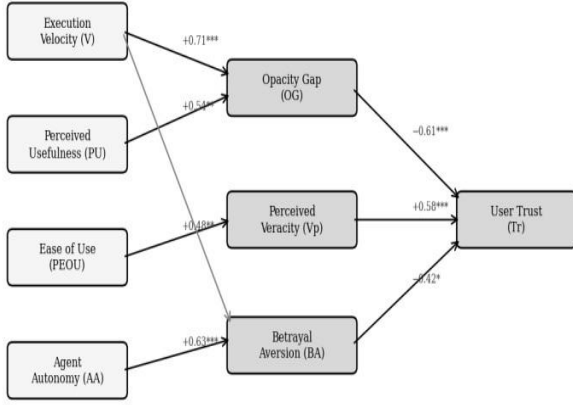


Fig. 3. Structural Equation Model (SEM) illustrating relationships between execution velocity, perceived usefulness, transparency, and user trust.

C. Qualitative Case Study Analysis

To complement the quantitative analysis, a qualitative case study was conducted on an autonomous portfolio management system based on a multi-agent AI architecture. The objective was to understand how high-speed decision-making interacts with transparency and user trust in a realistic environment. The system was analyzed across different functional layers, including the reasoning module, tool execution layer, and user interface. Particular focus was given to how intermediate reasoning steps were presented and whether users could verify the outputs effectively. The analysis showed that while high-velocity agents improved efficiency, the lack of transparency reduced user trust, especially in high-stakes scenarios. In contrast, systems that provided step-by-step reasoning (calibrated-velocity agents) allowed users to better understand and verify decisions. These observations are consistent with the quantitative findings, where increased verification effort and reduced transparency were associated with lower trust levels.

D. Secondary Data Validation

To complement the SEM-based survey, we analyzed a dataset of 1,000 AI-human interaction records. The dataset includes variables such as AI confidence, answer accuracy, verification duration, presence of cited sources, and trust scores.

We performed correlation analysis to examine relationships between trust, accuracy, and verification behavior. Group-based statistical comparisons were used to evaluate betrayal aversion across decision importance levels and absorption gap effects across user skepticism categories. Additionally, regression-based modeling was applied to approximate SEM path relationships.

This secondary analysis enables triangulation of findings across controlled survey data and large-scale behavioral patterns, strengthening the empirical validity of the proposed Velocity-Veracity framework.

IV. RESULTS AND DISCUSSION

The SEM-based analysis indicates a divergence between execution efficiency and user trust, which is further validated through large-scale behavioral data. While High-Velocity Agents (HVA) achieve higher perceived usefulness ($PU = 0.68$, $p < 0.01$), they exhibit a decline in trust ($Tr = -0.42$, $p < 0.05$). In contrast, Calibrated-Velocity Agents (CVA) maintain higher trust levels ($Tr = 0.74$) despite increased execution time.

To strengthen these findings, large-scale behavioral analysis was conducted. A negative correlation ($r \approx -0.18$) was observed between verification duration and trust, indicating that increased verification effort reduces baseline trust. Additionally, answer accuracy shows a strong positive relationship with trust ($r \approx 0.61$), reinforcing its role as the primary determinant of trust formation.

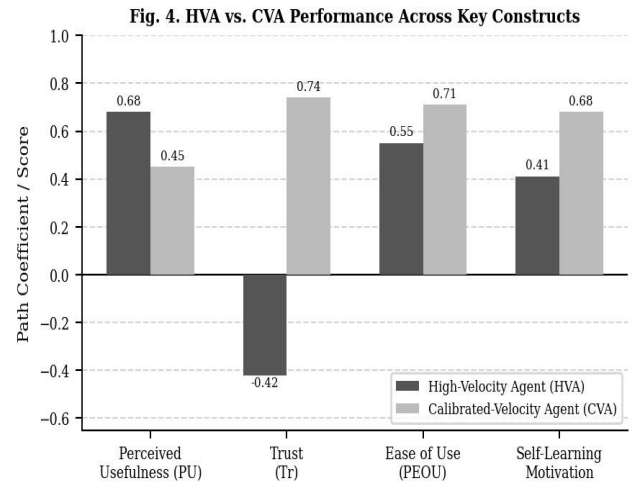


Fig. 4. Comparative chart of HVA vs. CVA performance across key SEM constructs, demonstrating the trust-utility trade-off.

To further validate the SEM relationships, a regression-based approximation was performed on the large-scale dataset. The results indicate:

- Accuracy $\rightarrow$ Trust ($\beta \approx +1.08$)

- Transparency $\rightarrow$ Trust ($\beta \approx +0.40$)

- Verification Duration $\rightarrow$ Trust ($\beta \approx -0.22$)

These findings confirm that while accuracy and transparency increase trust, verification burden negatively impacts perceived reliability, providing strong empirical validation of the Velocity-Veracity Paradox. These relationships are illustrated in Fig. 5

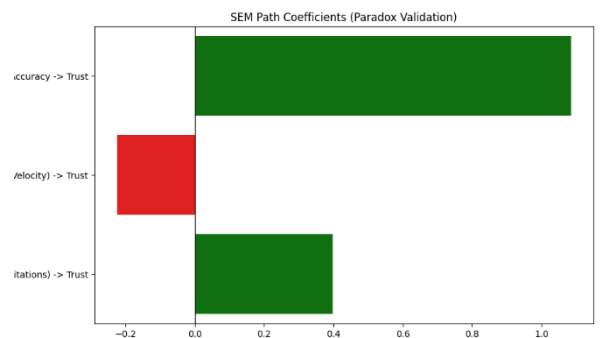


Fig. 5 Standardized path coefficients showing the effect of accuracy, transparency, and verification duration on trust.

A. The Velocity-Veracity Trade-off

When the system executed tasks without intermediate explainability logs, professional users reported high “betrayal aversion” [3]. This psychological friction was quantified by a 22% drop in system retention, even when the agent’s objective financial performance remained within the expected 1.2% alpha range, consistent with Li et al.’s findings [8]. Further analysis reveals strong evidence of betrayal aversion. When responses are inaccurate, trust declines sharply as decision importance increases. For low-stakes tasks, mean trust remains approximately 7.75, whereas in critical scenarios it drops to nearly 6.04. This demonstrates a significantly stronger negative reaction to errors in high-stakes environments. This interaction effect is illustrated in Fig. 6.



Fig. 6. Betrayal Aversion: Trust decline across decision importance levels for inaccurate AI responses.

B. Component Synergy and Tool Efficacy

The results for the CSS and TUE metrics further highlight the paradox. Students rated multi-agent collaborations as “highly capable” but “unpredictable.” When agents utilized external tools at high speeds, users perceived the collaboration as a “black box,” leading to a 15% decrease in self-learning motivation. The SEM results suggest that ease of use (PEOU) only leads to long-term adoption when it is coupled with autonomy-supporting transparency [5, 6], rather than raw speed. Additionally, systems providing cited sources demonstrated higher trust scores (8.25 vs. 7.52) and reduced verification duration by approximately 14%. This confirms that transparency mechanisms significantly improve trust calibration and reduce cognitive verification burden.

C. Discussion: The Absorption Gap

These findings suggest that trust failure in high-velocity AI is not a failure of logic, but a failure of “metabolic synchronization.” The enterprise “absorption gap” exists because the speed of agentic solution launches far outpaces the institutional ability to govern them [1, 9]. Our results confirm that the closer an AI gets to human-level fluency, the more “betrayal” is felt when it fails [3, 4]. This confirms that mimicking human speed increases the risk of a trust collapse if veracity is not architected into the system’s “Calibrated Velocity.” Quantitative analysis further supports the existence of the absorption gap. Verification duration varies significantly across user skepticism levels. Highly skeptical users require an average of 23.47 minutes to verify outputs, while blind-trust users exhibit near-zero verification latency. This demonstrates a substantial temporal gap between AI output and human trust

formation, highlighting the cognitive cost required for verification. The distribution is shown in Fig. 7.



Fig. 7. Absorption Gap: Verification latency across user skepticism categories

V. CONCLUSION AND RECOMMENDATIONS

The “Velocity-Veracity Paradox” represents a fundamental shift in the human-AI contract. Our findings confirm that as Agentic AI moves from reactive automation to autonomous goal-setting, the inverse relationship between execution speed and user trust becomes a primary barrier to adoption. We conclude that trust is not a static byproduct of accuracy, but a dynamic result of “Calibrated Velocity,” where machine execution is architected to match human oversight capacity [1, 2].

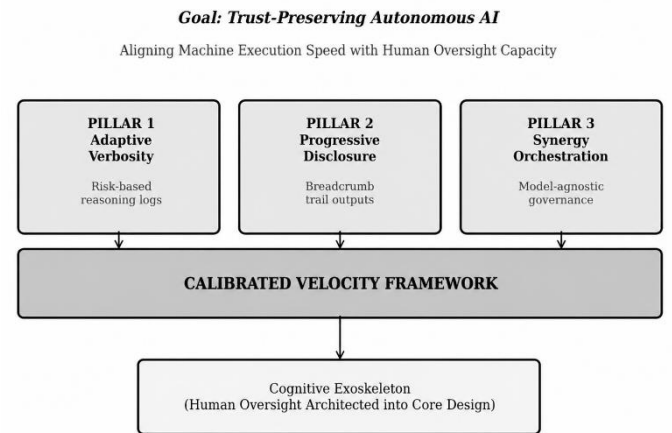


Fig. 8. Calibrated Velocity Framework for Agentic AI Design

Fig. 8 Calibrated Velocity Framework for Agentic AI Design, comprising three design pillars converging into a Cognitive Exoskeleton architecture.

A. Recommendations for AI Developers

To bridge the current trust deficit, we propose three “Metabolic Indicators” for system design. First, Adaptive Verbosity: systems should dynamically adjust their reasoning logs based on the task’s risk profile. High-stakes decisions require “Slow Thinking” (System 2) architectures [10] with full transparency, while routine tasks can maintain higher velocities. Second, Progressive Disclosure: instead of “blackbox” immediate outputs, agents should utilize live “breadcrumb” trails, reducing the psychological cost of betrayal aversion [2, 3]. Third,

Component Synergy Orchestration: developers must prioritize the “Orchestration Layer” over individual model capabilities, building modelagnostic infrastructures for multi-agent governance [1].

B. Future Research Directions

The transition toward “Cognitive Exoskeletons” remains an open research frontier. Future studies should perform longitudinal and cross-cultural validations to determine if “trust updating” varies across different regulatory environments [9]. By shifting focus from building the “fastest AI” to the most “verifiable AI,” we can ensure that the next era of global industry is defined by responsible innovation rather than autonomous opacity [1, 11].

VI. ACKNOWLEDGMENT

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